Screening Test
Sub-Junior

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test – Kaprekar Contest

NMTC at SUB JUNIOR LEVEL – VII & VIII Standards

Saturday, 27th August, 2016

Note:

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
2. Diagrams are only visual aids; they are NOT drawn to scale.
3. You are free to do rough work on separate sheets.
4. Duration of the test: 2 pm to 4 pm – 2 hours.

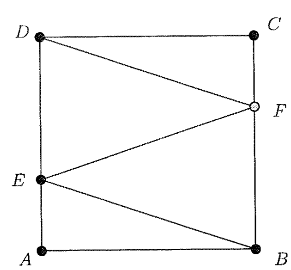
PART – A

Note

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{2}$ mark.**

1. On the square $ABCD$, point E lies on the side AD and F lies on BC so that $BE = EF = FD = 30\,\text{cm}$. The area of the square (in square cms) is

   A. 300 B. 900 C. 810 D. None of these



2. If $2016 = 2^x 3^y 5^z 7^u$ where x, y, z, u are non negative integers, the value of $x + y + 2016z + 3u$ is

   A. 10 B. 11 C. 12 D. 13

3. A is the area of triangle of sides 25, 25 and 30. B is the area of a triangle of sides 25, 25 and 40. Then

A. $A = B$ B. $A < B$ C. $A = 3B$ D. $A = 2B$

4. A number on being divided by 5 leaves a remainder 2 and when divided by 7, leaves a remainder 4. The remainder when the same number is divided by 5×7 is

A. 20 B. 23 C. 32 D. None of these

5. The sum of three numbers is 204. If the ratio of the first to second is $2:3$ and that of the second to third is $5:3$. Then the second number is

A. 60 B. 65 C. 58 D. 90

6. Anirud goes to the vegetable market to buy onions. He carries money to buy 3 kg onions. But on reaching the market he finds that the price of onions has increased by 20%. If he buys onions for the money he has, the quantity he could buy is

A. 2.5 kg B. 2.2 kg C. 2 kg D. 2.8 kg

7. Two runners cover the same distance at the rate of 15 km per hour and 16 km per hour respectively. The first runner took 16 minutes longer than the second to cover the distance. Then the distance (in km) is

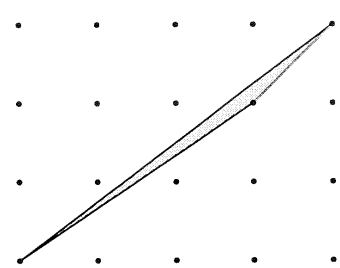
A. 58 B. 64 C. 66 D. 73

8. The total surface area of a cube is 600 cm^2. The length of its diagonal in cms is

A. 10 B. 30 C. $10\sqrt{2}$ D. $10\sqrt{3}$

9. In the figure below the distance between any two horizontal or vertical dots is one unit. The area of the triangle shown is

A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{1}{6}$



10. The number of natural numbers less than 400 that are not divisible by 17 or 23 is

A. 290 B. 320 C. 360 D. 370

11. If a is a % of b and b is b% of c, where a is a positive real number, then the value of c is

A. 120 B. 200 C. 150 D. 100

12. The sum of the reciprocals of all the positive integers that divide 24 is

A. $\frac{7}{2}$ B. $\frac{1}{2}$ C. $\frac{5}{2}$ D. $\frac{3}{2}$

13. N is a positive integer and p, q are primes. If $N = pq$ and $\frac{1}{N} + \frac{1}{p} = \frac{1}{q}$, the value of N is

A. 6 B. 7 C. 8 D. 9

14. The number of two digit numbers that leave a remainder 1 when divided by 4 is

A. 20 B. 21 C. 22 D. 23

15. If $(a+b)^2 + (b+c)^2 + (c+d)^2 = 4(ab + bc + cd)$, then

A. $a+b$ or $b+c$ or $c+d$ must be zero

B. Two of a, b, c, d are zero and other two non zero

C. $a = b = c = d$

D. None of these

PART – B

Note

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{4}$ mark.**

16. A race horse eats $(3a + 2b)$ bags of oats every week. The number of weeks in which it can eat $(12a^2 - 7ab - 10b^2)$ bags of oats is ———

17. Choose 4 digits a, b, c, d from $\{2, 0, 1, 6\}$ and form the number $(10a + b)^{10c+d}$. For example, if $a = 2, b = 0, c = 1, d = 6$, we will get 20^{16}. For all such choices of a, b, c, d the number of distinct numbers that will be formed is ———

18. A fraction F becomes $\frac{1}{2}$ when its denominator is increased by 4 and becomes $\frac{1}{3}$ when its numerator is decreased by 5. Then F equals ———

19. The average of 5 consecutive positive integers starting with m is n. Then the average of 5 consecutive integers starting with n is (in terms of m) is ———

20. Two boys came to Mahadevan and asked his age. Mahadevan told, "Delete all the vowels and repeated letters from my name. Find the numerical value of the remaining letters (for example, D has value 4, G has 7 etc). Add all of them. Find the number got by interchanging its digits. Add both the numbers. That is my age". One boy ran away. The other boy calculated correctly. The age of Mahadevan is ———

21. If

$$A = \frac{2^4 + 2^4}{2^{-4} + 2^{-4}}, \quad B = \frac{3^2 + 3^2}{3^{-2} + 3^{-2}}, \quad C = \frac{4^2 + 4^2}{4^{-2} + 4^{-2}}$$

the integral part of $\frac{A+C}{B}$ is ———

22. A six-digit number is formed using the digits 1, 1, 2, 2, 3, 3. The number of 6-digit numbers in which the 1s are separated by one digit, 2s are separated by two digits and 3s are separated by 3 digits is ———

23. In the addition shown below, P, I, U are digits. The value of U is ———

$$PI + PI + PI + PI = UP$$

24. There are four cows, eight hen, a fish, a crow, a girl and a boy in a garden. Outside the garden there is one dog, a peacock and some cats. The number of legs of all of them inside the garden is equal that outside the garden. The number of cats is ———

25. Two sides of a triangle are 8 cm and 5 cm. The length of the third side in cms is also an integer. The number of such triangles is ———

26. If $a^2 - a - 10 = 0$, then $(a+1)(a+2)(a-4)$ is ———

27. There are 5 points on the circumference of a circle. The number of chords which can be drawn joining them is ———

28. Each side of an equilateral triangle is 3 cm longer than each side of a square. The total perimeter of the square and the triangle is 51 cm. Then the side of the triangle in cms is ———

29. The largest three digit number that is a multiple of 3 and 5 is ———

30. Consider the sequence $0, 6, 24, 60, 120, \ldots$. The 6th term of this sequence is ———

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